

Wealthfront Investment Methodology White Paper

How to build and manage a diversified investment portfolio

Introduction

Wealthfront aims to deliver an automated investment management service that maximizes the net-of-fee, after-tax, real investment return for each client's particular tolerance for risk. This paper describes the methodology we employ to achieve this objective.

Our investment management team, led by Dr. Burton Malkiel, renowned economist and author of *A Random Walk Down Wall Street*, designed a service that starts with a portfolio diversified across relatively uncorrelated asset classes, customized for your particular risk tolerance. We invest with an equity orientation to maximize **long-term** returns. Each of our selected asset classes is represented by a low cost, passive ETF. We continuously monitor and periodically rebalance your portfolio to maximize your chance of investment success for the long run. We also attempt to minimize your taxes by analyzing the taxes likely to be generated by any given asset class, and then allocating different asset classes in taxable and non-taxable (retirement) portfolios.

We use Modern Portfolio Theory (MPT) to identify the ideal portfolio for each client. The economists who developed MPT, Harry Markowitz and William Sharpe, received the Nobel Prize in Economics in 1990 for their groundbreaking research. Today, MPT is the most widely accepted framework for managing diversified investment portfolios. MPT has its limitations, especially in the area of very low probability significant downside scenarios, but we and our advisors believe it is the best framework on which to build a compelling investment management service.

Sophisticated investment management services were previously available only to wealthy investors through financial advisors. Typically, those advisors charge annual management fees in excess of 1%, and require account minimums of at least \$1 million. By implementing a completely software-based solution, informed by world-class financial expertise, Wealthfront is able to deliver its automated investment management service at much lower cost. We democratize access to high-quality financial advice.

Our investment methodology employs five steps:

- Identify an ideal set of asset classes for the current investment environment
- Select low cost ETFs to represent each asset class
- Determine your risk tolerance to create the appropriate portfolio for you
- Apply Modern Portfolio Theory to allocate among the chosen asset classes for your risk tolerance
- Monitor and periodically rebalance your portfolio

Finding Asset Classes

Research consistently has found the best way to maximize returns across every level of risk is to combine asset classes rather than individual securities (Markowitz, 1952; Sharpe, 1964; Brinson, Hood & Beebower, 1986; Brinson, Singer & Beebower, 1991; Ibbotson & Kaplan, 2000). Therefore the first step in our methodology is to identify a broad set of diversified asset classes to serve as the building blocks for our portfolios. We consider each asset class's long-term historical behavior in different economic scenarios, risk-return relationship conceptualized in asset pricing theories, and expected behavior going forward based on long-term secular trends and the macroeconomic environment. We also evaluate each asset class on its potential for capital growth and income generation, volatility, correlation with the other asset classes (diversification), inflation protection, cost to implement via ETF and tax efficiency.

Asset classes fall under three broad categories: stocks, bonds and inflation assets. Stocks, despite their high volatility, give investors exposure to economic growth and offer the opportunity for long-term capital gains. Stocks provide effective long-run inflation protection and are relatively tax efficient due to the favorable tax treatment on long-term capital gains and stock dividends (relative to the way ordinary income is taxed). Bonds and bond-like securities are the most important income-producing asset classes for income-seeking investors. Although bonds have lower return expectations, they provide a cushion for stock-heavy portfolios during economic turbulence due to their low volatility and low correlation with stocks. Most bonds are tax inefficient because bond interest income is taxed at ordinary income tax rates, except tax-exempt Municipal Bonds. Assets that protect investors from inflation in both moderate and high inflation environments include Treasury Inflation-Protected Securities (TIPS), Real Estate and Natural Resources. Their prices tend to be highly correlated with inflation.

Based on a thorough analysis, our investment team currently works with the asset classes listed in Table 1.

Table 1: Asset classes and their functions

Asset class	Benefits
US Stocks	Capital growth, long-run inflation protection, tax efficiency
Foreign Developed Stocks	Capital growth, long-run inflation protection, tax efficiency
Emerging Market Stocks	Capital growth, long-run inflation protection, tax efficiency
Dividend Growth Stocks	Capital growth, income, long-run inflation protection, tax efficiency
US Government Bonds	Income, low historical volatility, diversification
Corporate Bonds	Income, low historical volatility, diversification
Emerging Market Bonds	Income, diversification
Municipal Bonds	Income, low historical volatility, diversification, tax efficiency
Treasury Inflation-Protected Securities (TIPS)	Income, low historical volatility, diversification, inflation protection
Real Estate	Income, diversification, inflation protection
Natural Resources	Diversification, inflation protection, tax efficiency

U.S. Stocks represent an ownership share in U.S.-based corporations. The U.S. has the largest economy and stockmarket in the world. Although the U.S. economy was hit hard in the 2008-2009 Financial Crisis and its pace of growth in the future is expected to slow compared with its historical growth rate, the U.S. economy is still one of the most resilient and active in the world, powered as it is by a remarkable innovation engine.

Foreign Developed Market Stocks represent an ownership share in companies headquartered in developed economies like Europe, Australia and Japan. Although the economies of Europe and Japan have experienced many struggles in the last two decades, Foreign Developed Markets represent a significant part of the world economy.

Emerging Market Stocks represent an ownership share in foreign companies in developing economies such as Brazil, China, India, South Africa and Taiwan. Compared with developed countries, developing countries have younger demographics, expanding middle classes and faster economic growth. They account for half of world GDP and that portion is likely to increase as the Emerging Markets develop. Emerging Market Stocks are more volatile, but we expect them to deliver higher returns than U.S. Stocks and Foreign Developed Markets Stocks for the long term.

Dividend Growth Stocks represent an ownership share in U.S. companies that have increased their dividend payout each year for the last ten or more consecutive years. They tend to be large-cap well-run companies in less cyclical industries and thus are less volatile than stocks more generally. Many companies in this asset class have higher dividend yields than their corporate bond yields and the yields on U.S. government bonds. In the current low interest rate environment, Dividend Growth Stocks emerge as an asset class that offers an income stream and capital growth potential.

U.S. Government Bonds are debt issued by the U.S. federal government and agencies to fund various spending programs. U.S. Government Bonds provide steady income, low historical volatility and low correlation with stocks. U.S. Government Bonds currently offer historically low yields and are expected to produce barely positive or even negative real returns due to the low interest rate policy currently administered by the Federal Reserve.

Corporate Bonds are debt issued by U.S. corporations with investment-grade credit ratings to fund business activities. They offer higher yields than U.S. Government Bonds due to higher credit risk, illiquidity and callability. In contrast to the U.S. government, most U.S. companies have gone through a deleveraging process and strengthened their balance sheets over the last few years.

Emerging Market Bonds are debt issued by governments and quasi-government organizations from emerging market countries. They offer higher yields than developed market bonds. Emerging Market Bonds had serial defaults in the 1980s, 1990s and even 2000s. However, the world has changed. Investors today worry more about potential defaults from developed market bonds rather than emerging market bonds. Emerging market countries, with younger demographics, stronger economic growth, healthier balance sheets and lower debt-to-GDP ratios, have less risk than most investors realize with respect to borrowing money.

Municipal Bonds are debt issued by U.S. state and local governments. Unlike most other bonds, Municipal Bonds' interest is exempt from federal income taxes. They provide individual investors in high tax brackets a tax efficient way to obtain income, low historical volatility and diversification.

Treasury Inflation-Protected Securities (TIPS) are inflation-indexed bonds issued by the U.S. federal government. Unlike nominal bonds, TIPS' principal and coupons are adjusted periodically based on the Consumer Price Index (CPI). Although TIPS currently have historically low yields, their inflation-indexed feature and low historical volatility makes them the only asset class that can provide income generation and inflation protection to risk averse investors.

Real Estate is accessed through publicly traded U.S. real estate investment trusts (REITs) that own commercial properties, apartment complexes and retail space. They pay out their rents as dividends to investors. REITs provide income, inflation protection and diversification benefits.

Natural Resources reflect the prices of energy (e.g., natural gas and crude oil). Natural Resources provide inflation protection and diversification. Investing in Natural Resources via exchange-traded products is also relatively tax efficient due to the favorable tax treatment on long-term capital gains and stock dividends.

The asset classes we deploy may evolve somewhat over time, depending on long-term macroeconomic factors and the evolution of ETFs in the marketplace.

Once we decide on our asset classes, our next step is to identify their optimal mix for each level of risk and type of account (taxable vs. retirement).

Allocating Assets

Mean-Variance Optimization

Wealthfront determines the optimal mix of our chosen asset classes by solving the "Efficient Frontier" using Mean-Variance Optimization (MVO) (Markowitz, 1952), the foundation of Modern Portfolio Theory. The Efficient Frontier represents the portfolios that generate the maximum return for every level of risk. Each portfolio is created by choosing a particular mix of asset classes that maximizes the expected return for a specific level of risk (as measured by variance), or equivalently minimizes the risk for a specific expected return. MVO calculates the best risk-return tradeoff when combining the asset classes in portfolios.

In addition to portfolio construction, we also use MVO as an important quantitative tool to evaluate how many asset classes we should use in a portfolio. If adding an asset class to the mix raises the efficient frontier, then it improves the risk-return tradeoff of the portfolios, (i.e. it offers a higher return for the same risk level or lower risk for the same return level).

MVO provides a powerful mathematical framework for evaluating portfolio risk-return tradeoffs. As you will see later in this paper, we also apply other quantitative approaches and qualitative assessment when choosing portfolios to manage.

Capital Market Assumptions

MVO requires, as inputs, estimates for each asset class's standard deviation, correlation and expected return.

To estimate each asset class's standard deviation (volatility), we consider its long-term historical standard deviation, its short-term standard deviation, and the expected volatility implied by its pricing in the options markets. Long-term historical estimates benefit from a larger sample size, short-term estimates capture market evolution and the option markets imply forward-looking volatility. To estimate correlation, we consider long-term historical correlation and short-term correlation.

Table 2 presents the correlations among the asset classes and Table 3 presents the standard deviation for each asset class.

The correlations between stocks and bonds remain low, confirming the benefit of diversifying with bonds. Correlations among different types of stocks have increased over the last few years. Foreign Developed Stocks and Emerging Market Stocks historically have been good diversifiers for U.S. Stocks, but this has not been as true recently. We use non-US stocks primarily for their return potential. Real Estate and Natural Resources are more correlated with stocks today than in the 1980s and 1990s, but still offer moderate diversification benefits. Emerging Market Bonds' recent historical volatility and implied volatility are much lower than they were in the 1980s and 1990s, reflecting the maturing of the asset class.

To estimate each asset class's expected returns, we start with the Capital Asset Pricing Model (CAPM) (Sharpe, 1964) as the baseline estimate. CAPM derives expected returns in market equilibrium under certain assumptions, and states that the expected return of an asset class is dictated by its systematic risk as measured by beta. Riskier asset classes command higher expected returns. Both MVO and CAPM are important constituents of Modern Portfolio Theory (MPT). We also form views on long-term return expectations for each asset class based on interest rates, credit spreads, dividend yields, GDP growth and other macroeconomic variables. We use the Black-Litterman model (Black & Litterman, 1992) and the Gordon growth model (Gordon, 1959) to adjust the CAPM returns with our views. We subtract ETF expenses from the gross return of each asset class to estimate its net-of-fee expected return. We also subtract the estimated tax liability due on each asset class's return to derive a net-of-fee, after-tax expected return. All returns are then input into the MVO model net of inflation (i.e. as "real" returns). Please see Table 4 for the details of how we calculate each asset class's expected returns. Our asset classes' expected returns are low compared to historical standards, primarily due to the low interest rate and slow economic growth environment. Note that expected returns are presented as real returns (adjusted for 2% estimated inflation) rather than nominal returns.

MVO is sensitive to input parameters and tends to produce concentrated and unintuitive portfolios if the parameters are naively specified. To overcome the difficulty of applying MVO in practice, Fischer Black and Robert Litterman proposed the Black-Litterman model while working at Goldman Sachs (Black & Litterman, 1992). Their model applies a technique that derives expected return parameters from equilibrium returns and manager views. It largely mitigates the optimizer's sensitivity problem and enables it to

produce diversified and intuitive portfolios. In addition, the Black-Litterman model provides a flexible framework to express views about asset class returns, which ultimately will be reflected in the asset allocation.

Table 2: Asset class correlation assumptions

US Stocks	1.0	0.9	0.8	0.8	0.7	0.7	0.0	-0.1	0.1	0.5	-0.2
Foreign Stocks	0.9	1.0	0.8	0.7	0.7	0.7	0.0	-0.1	0.2	0.5	-0.1
Emerging Markets	0.8	0.8	1.0	0.6	0.6	0.7	0.1	-0.1	0.2	0.6	-0.1
Dividend Stocks	0.8	0.7	0.6	1.0	0.7	0.6	0.0	0.0	0.1	0.6	-0.1
Real Estate	0.7	0.7	0.6	0.7	1.0	0.6	0.1	0.0	0.3	0.4	0.0
Natural Resources	0.7	0.7	0.7	0.6	0.6	1.0	0.0	-0.3	0.0	0.5	-0.3
TIPS	0.0	0.0	0.1	0.0	0.1	0.0	1.0	0.5	0.7	0.3	0.7
Municipal Bonds	-0.1	-0.1	-0.1	0.0	0.0	-0.3	0.5	1.0	0.6	0.1	0.7
Corporate Bonds	0.1	0.2	0.2	0.1	0.3	0.0	0.7	0.6	1.0	0.5	0.8
Emerging Market Bonds	0.5	0.5	0.6	0.6	0.4	0.5	0.3	0.1	0.5	1.0	0.2
US Government Bonds	-0.2	-0.1	-0.1	-0.1	0.0	-0.3	0.7	0.7	0.8	0.2	1.0

CORRELATION



Table 3: Asset class standard deviation assumptions

Asset Class	Standard Deviation (Annualized)
US Stocks	16%
Foreign Developed Stocks	18%
Emerging Market Stocks	24%
Dividend Growth Stocks	14%
US Government Bonds	5%
Corporate Bonds	5%
Emerging Market Bonds	7%
Municipal Bonds	5%
TIPS	5%
Real Estate	18%
Natural Resources	22%

Table 4: Asset class expected real returns

Note: long-term inflation expectation is 2%.

Asset class	CAPM Return	Wealthfront View	Black-Litterman Gross Return	Net-of-Fee Return	After-Tax Net-of-Fee Return
US Stocks	5.3%	4.3%	4.6%	4.5%	3.9%
Foreign Stocks	6.2%	3.3%	5.3%	5.2%	4.4%
Emerging Markets	8.1%	5.4%	7.0%	6.8%	5.9%

Dividend Stocks	3.7%	5.3%	3.5%	3.4%	2.5%
US Government Bonds	-0.8%	-1.5%	-0.9%	-1.0%	-1.4%
Corporate Bonds	-0.2%	1.0%	-0.3%	-0.4%	-1.1%
Emerging Market Bonds	1.0%	2.0%	1.1%	0.8%	-0.5%
Municipal Bonds	-0.8%	1.0%	-0.7%	-0.9%	-1.0%
TIPS	-0.5%	-2.3%	-0.7%	-0.8%	-1.2%
Real Estate	5.0%	3.5%	4.4%	4.3%	2.7%
Natural Resources	6.2%	3.0%	4.7%	4.6%	3.9%

We update our estimates annually which likely results in small changes to our recommended asset allocations. Existing clients' portfolios are rebalanced to account for the new estimates if the changes in estimates leads a particular asset class percentage to fall outside the thresholds described in the rebalancing section below.

Portfolio Construction

In addition to estimating parameters carefully for MVO, we enforce minimum and maximum allocation constraints for each asset class. This method is widely used to ensure proper portfolio diversification, mitigate parameter estimation errors and express investor preferences. Table 5 shows the minimum/maximum allocation constraints we chose for each of the asset class. We selected 5% as a minimum allocation because anything less than that does not provide meaningful diversification benefits in our estimation. We selected 35% as the maximum allocation to ensure sufficient diversification from meaningful allocations to the other asset classes. Other sources including ([Swensen, 2005](#)) recommend similar min and max allocations by asset class. Note that we do not enforce a minimum allocation to TIPS because they are not efficient for investors with moderate to high risk tolerance.

Table 5: Asset class allocation constraints

Asset Class	Minimum Allocation	Maximum Allocation
US Stocks	5%	35%
Foreign Developed Stocks	5%	35%
Emerging Market Stocks	5%	35%
Dividend Growth Stocks	5%	35%
US Government Bonds	5%	35%
Corporate Bonds	5%	35%
Emerging Market Bonds	5%	35%
Municipal Bonds	5%	35%
TIPS	0%	35%
Real Estate	5%	35%
Natural Resources	5%	35%

The different ways in which the source of each asset class's likely return is taxed plays an important role in determining whether an asset class is appropriate for a taxable account, retirement account or both. Table 6 displays the tax efficiency of the asset classes.

Table 6: Asset class relative tax efficiency

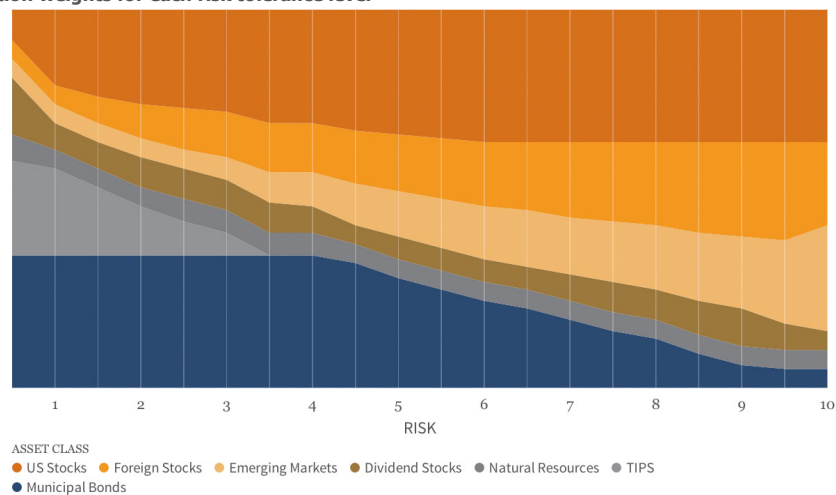
Asset Class	Tax Efficiency
US Stocks	Efficient
Foreign Developed Stocks	Efficient
Emerging Market Stocks	Efficient
Dividend Growth Stocks	Efficient
US Government Bonds	Inefficient
Corporate Bonds	Inefficient
Emerging Market Bonds	Inefficient
Municipal Bonds	Efficient

TIPS	Inefficient
Real Estate	Inefficient
Natural Resources	Efficient

In our MVO framework we found that our allocations, outlined below, were insensitive to varying tax assumptions. More specifically, the allocations were robust across the top four federal tax brackets and various state income tax assumptions.

Exhibit 1 presents the optimal asset allocations solved using the parameters described above for taxable accounts. Seven of our 11 possible asset classes were tax efficient enough to be deployed in our taxable allocation – TIPS, Municipal Bonds, Dividend Growth Stocks, US Stocks, Foreign Developed Stocks, Emerging Market Stocks and Natural Resources. As the risk level increases from left to right, allocation to conservative asset classes such as TIPS and Municipal Bonds decreases, while allocation to aggressive asset classes such as US Stocks, Foreign Developed Stocks, Emerging Market Stocks increases. Dividend Growth Stocks fall somewhere between conservative and aggressive asset classes. Municipal Bonds emerge as the primary bond asset class in the allocation because they have higher net-of-fee, after-tax expected returns due to their federal tax exemption. The Muni bond ETF we use is only exempt from federal taxes. We haven't yet found state specific Muni ETFs that are exempt from state income taxes that have sufficient liquidity to be included in our asset allocation. We will continue to monitor the market and plan on adding them as soon as they become practical. TIPS, although tax inefficient, still appear in conservative portfolios because they are the only low-volatility asset class offering inflation protection. All types of stocks remain in the allocation because stock dividends are taxed at qualified dividend tax rates, which are less than ordinary income tax rates. Natural Resources emerge due to their tax efficiency. Real Estate, Corporate Bonds and Emerging Market Bonds fall out because their dividends or interest are taxed at ordinary income tax rates, which make them tax inefficient.

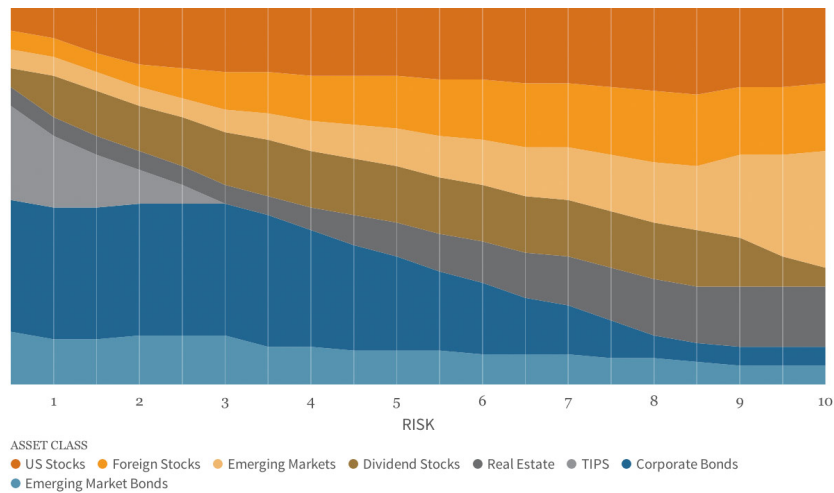
Exhibit 1: Taxable allocation weights for each risk tolerance level



It is important to note that we did not consider the benefits from tax-loss harvesting when designing our portfolio allocations for our base level service. For more information on our tax-loss harvesting service, which is available to clients with a taxable account, please see <https://www.wealthfront.com/tax-loss-harvesting>.

Exhibit 2 presents the optimal asset allocations solved using the parameters described above for retirement accounts. We evaluated 11 asset classes and chose to employ eight (TIPS, Corporate Bonds, Emerging Market Bonds, Dividend Growth Stocks, US Stocks, Foreign Developed Stocks, Emerging Market Stocks and Real Estate) in our retirement portfolio allocation based on our MVO framework. Similarly, as the risk level increases from left to right, allocation to conservative asset classes such as TIPS and Corporate Bonds decreases, while allocation to aggressive asset classes such as U.S. Stocks, Foreign Developed Stocks, Emerging Market Stocks and Real Estate increases. Emerging Market Bonds and Dividend Growth Stocks behave somewhere between conservative and aggressive asset classes. TIPS are allocated only in the conservative portfolios for risk-averse investors, while risk tolerant investors have larger allocations to stocks and Real Estate for inflation protection. U.S. Government Bonds, Municipal Bonds and Natural Resources are not used because they don't add economic benefit (i.e. increased return for the same risk) in the presence of the other eight asset classes.

Exhibit 2: Tax-free allocation weights for each risk tolerance level

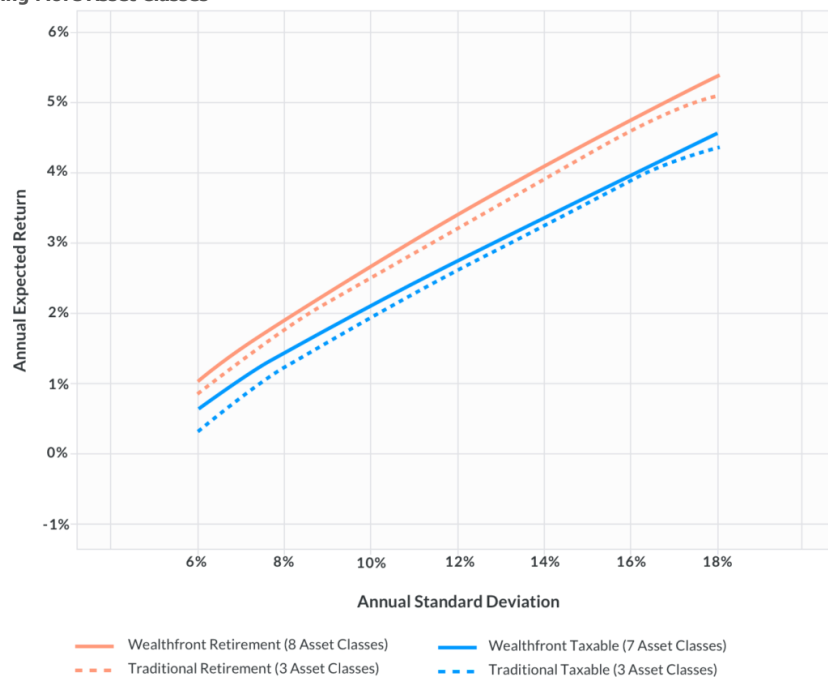


How Many Asset Classes?

Traditionally, financial advisors allocated their client portfolios across three asset classes (U.S. Stocks, Foreign Developed Stocks and U.S. Government Bonds). Thanks to MPT, we can compare a traditional three-asset class allocation with the Wealthfront seven and eight asset class allocations. Exhibit 3 illustrates the benefit of adding more asset classes. Adding more uncorrelated asset classes to the traditional three asset class allocation raises the Efficient Frontier by approximately 0.10% – 0.20% per year for retirement accounts and by approximately 0.05% – 0.30% per year for taxable accounts. In other words, adding more asset classes increases real return for each risk level, or reduces risk for each return level. Missing out on the additional asset classes represents a substantial opportunity cost for investors.

There is no definitive answer to the question “how many asset classes investors should hold?” It is relatively easy to improve the risk-return tradeoff of a two or three asset class portfolio. It gets increasingly difficult to improve a portfolio already diversified across seven or eight asset classes. Going beyond a certain level of complexity generally reaches diminishing returns, especially when you incorporate ETF costs into your decision-making. Having said that, we will continue to evaluate new relatively uncorrelated asset classes that can be implemented using low-cost liquid ETFs, to improve our asset allocation.

Exhibit 3: Benefit of Adding More Asset Classes



Note: Traditional retirement and taxable asset allocations consist of three asset classes (US Stocks, Foreign Developed Stocks and US Government Bonds). Wealthfront retirement allocation consists of eight asset classes (US Stocks, Foreign Developed Stocks, Emerging Market Stocks, Dividend Growth Stocks, Corporate Bonds, Emerging Market Bonds, TIPS and REITs). Wealthfront taxable allocation consists of seven asset classes (US Stocks, Foreign Developed Stocks, Emerging Market Stocks, Dividend Growth Stocks, Municipal Bonds, TIPS and Natural Resources). Y-axis represents expected annual real return, net of ETF expense ratios; rebalancing is not assumed; results are based on forward-looking analysis done by Wealthfront. Long-term inflation expectation is 2%.

Handling Small Accounts

Wealthfront accounts can be as small as \$500, which doesn't always provide sufficient cash for meaningful exposure to the 6 to 8 asset classes we typically recommend.

As a result, for such small accounts, we use a process of holistic optimization to select the available investment ETFs that best match the expected performance of the desired portfolio allocation while minimizing the “cash drag” from any uninvested assets. Our backtesting shows that this optimization process typically results in portfolios with a smaller number of asset classes, minimal cash drag, and a strong match for the historical performance of the desired target portfolio. What’s more, as these accounts grow in size they gracefully evolve into our typical 6 to 8 asset portfolio allocations.

Selecting Investment Vehicles

Wealthfront uses cost-effective, index-based Exchange Traded Funds (ETFs) to represent each asset class. In contrast, many financial advisors have historically recommended actively managed mutual funds. Mutual funds were convenient because they could be chosen easily using a well-known ratings system offered by Morningstar. In 2010, Morningstar admitted its ratings system did not successfully identify mutual funds that could outperform the market in the future (Kinnel, 2010). Not surprisingly, a significant amount of research has been published that shows the majority of mutual funds (65-75%) underperform the market (Bogle, 2009; Malkiel, 2012) and those that outperform in one period are unlikely to outperform in subsequent periods. A widely cited paper on the subject showed mutual funds underperformed the Vanguard S&P 500® index fund by an average of 2.1% per year pre-tax over a 20-year period due to high fees and poor stock selection (Arnott, Berkin, & Ye, 2000).

As a result, index funds and more specifically passive index ETFs have exploded over the past 10 years. More than 1,500 ETFs have been created and in aggregate, ETFs have accumulated assets of more than \$1 trillion. Unlike mutual funds, ETFs do not have a standard rating agency, which has made it difficult for the average investor (or advisor) to understand ETF costs and determine which are the best way to “play” each asset class.

Wealthfront periodically reviews the entire population of ETFs to identify the most appropriate ones to represent each of its six recommended asset classes. We look for ETFs that minimize cost and tracking error, offer ample market liquidity, and minimize the lending of their underlying securities.

Most investors are surprised to learn that ETFs do not exactly track the indices they were created to mimic. The higher the variance (tracking error) from its selected benchmark, the less appropriate an ETF is to represent its asset class. An ETF issuer can reduce its tracking error by improving its operational systems, but that adds expense which is typically passed on as a higher management fee to the investor. In other words, expense and tracking error are often inversely correlated. We pay careful attention to this trade-off.

We choose ETFs that are expected to have sufficient liquidity to allow withdrawals at any time. Newly issued ETFs usually take a while before they are appropriate for recommendation.

In addition, most investors do not realize that many ETF issuers generate income from lending out their underlying securities to hedge funds to enable short sales; the more prevalent the lending, the higher the risk to the ETF buyer. We prefer ETFs that either minimize lending or share the lending revenue with their investors to lower management fees.

We attempt to choose ETFs that are *ideal in the context of an entire portfolio*. We are far more interested in the impact each ETF has on a portfolio’s overall net of fee, after-tax, risk-adjusted return than how it may be evaluated on its own. For example we chose Vanguard’s Total Stock Market ETF (Ticker: VTI) to represent US Stocks because it provides a broad market exposure to large, mid and small capitalization stocks in a more optimal net of fee, after-tax, risk-adjusted fashion than breaking the US Stock allocation into three separate market cap focused ETFs. However we may choose a particular ETF that has a much higher management fee than alternatives if its superior anticorrelation with the other asset classes results in a higher overall net of fee, after-tax, risk-adjusted return.

Determining Your Risk

Once the Efficient Frontier has been established, it is necessary to pinpoint an investor’s risk tolerance in order to identify the ideal asset allocation for her needs. Rather than asking the typical 25 questions asked by financial advisors to identify an individual’s risk tolerance, Wealthfront combed behavioral economics research to simplify our risk identification process to only a few questions. For example, we are able to project an individual’s income growth and savings rate based on her age and current income. We ask prospective clients questions to evaluate both their objective *capacity* to take risk and subjective *willingness* to take risk. Our view is that sophisticated algorithms can do a better job of evaluating risk than the *average* traditional advisor.

We ask subjective risk questions to determine both the level of risk an individual is willing to take and the consistency among her answers. The less consistent the answers, the exponentially less risk tolerant the investor is likely to be. For example, if an individual is willing to take a lot of risk in one case and very little in another, then she is inconsistent and is therefore assigned a lower risk tolerance score than the simple weighted average of her answers.

We ask objective risk questions to estimate with as few questions as possible whether the individual is likely to have enough money saved at retirement to afford her likely spending needs. The greater the excess income, the more risk the customer is able to take. Conversely if her expected retirement income is less than her likely retirement spending needs, then she cannot afford to take much risk with her investments.

Our overall risk metric combines subjective and objective risk tolerance with a heavier weighting to whichever component is more risk averse. We chose this approach because behavioral economics research shows individuals consistently overstate their true risk

tolerance, especially male investors who are educated and overconfident (Barber & Odean, 2001). Relying on an investor's biased answers may lead to a more volatile portfolio than appropriate, which could increase the likelihood the investor sells when the market declines. DALBAR published an often-quoted study that observed the average equity investor underperformed S&P 500 by 4.32% on an annualized basis during the 20-year period 1992-2011 due to consistently buying after the market has risen and selling when the market declines (DALBAR, 2012).

We email our clients quarterly to determine if anything in their financial profile has changed that may affect their risk tolerance. For example getting married, having kids, benefiting from equity appreciation associated with an IPO or being promoted to a significantly higher paying job can have a major impact on the risk score we apply and therefore one's ideal investment mix. In addition we gradually adjust clients' investment mixes as they age to make sure they have less volatility as their retirement approaches.

We allow clients to adjust their assigned risk score once every 30 days, in the event they want a more or less conservative allocation based on their individual circumstances. We warn them in advance that it might not be appropriate for their ultimate goals. We restrict the risk score to be changed only once every 30 days as we don't believe it should be used as a market timing tool. We also may limit the number of times a client can change her risk score in order to further discourage attempts to time the market. Wealthfront discourages market timing because we believe attempting to time the market is one of the most serious mistakes an individual investor can make.

We select a portfolio for each client on the Efficient Frontier by maximizing the following classic utility function popularized by Nobel Prize winner Harry Markowitz, parameterized with our risk tolerance metric:

$$\text{Max}(r, \sigma) \quad r - \sigma^2 / 2\tau$$

where:

- r denotes a portfolio's expected return,
- σ denotes a portfolio's standard deviation,
- τ maps to Wealthfront's risk tolerance metric, which is calibrated to a scale of 0 – 10, where 5 corresponds to the market portfolio and the asset classes are in proportion to global capital markets.

This utility function measures an investor's happiness with her portfolio. It is assumed an investor prefers to find the optimal balance between return and risk while maximizing expected return and minimizing standard deviation. If an investor has a relatively high risk tolerance, she will focus on maximizing returns and will land on the high end of the Efficient Frontier. In the case of low risk tolerance, she will focus on minimizing risk and will land on the low end of the Efficient Frontier.

Alternatively, we could formulate the MVO as follows:

$$\text{Max}_w \quad \mu^T w - \frac{1}{2} w^T \Sigma w \quad \text{s.t.} \quad 1^T w = 1, a \leq w \leq b$$

where:

- μ denotes the asset class expected returns,
- Σ denotes the asset class covariance matrix,

- w denotes the asset class weights,
- τ maps to Wealthfront's risk tolerance metric,
- a and b are the minimum and maximum allocation constraints.

Note the constraints dictate that the asset class weights sum to one. We only consider long-only portfolios and also enforce the minimum and maximum constraints on the weights.

Exhibit 4 represents a Wealthfront investment recommendation for a taxable account worth \$100,000 with a risk tolerance score of 7 on a scale of 0 to 10, where 0 is the least risk tolerant, 10 is the more risk tolerant, and 7 is the average risk tolerance score among our clients. Exhibit 5 represents a Wealthfront investment recommendation for a retirement account worth \$100,000 with a risk tolerance score of 7. Both allocations are heavy in stocks and appropriate for risk tolerant investors.

Exhibit 4: Wealthfront investment recommendation for a taxable account



Exhibit 5: Wealthfront investment recommendation for a retirement account



Rebalancing and Ongoing Monitoring

A portfolio created using MPT-based techniques will not stay optimized over time. The composition of any investment portfolio will naturally drift as capital markets move and certain holdings outperform others. This typically results in two adverse outcomes: (1) portfolio risk increases as the equity portion of the portfolio grows beyond its original allocation, and (2) allocations become sub-optimally mixed. To maintain the intended risk level and asset allocations, a portfolio must be periodically rebalanced back to its original targets. Sophisticated algorithms are required to optimize rebalancing subject to tax and trading expense effects.

Wealthfront monitors our clients' portfolios and periodically rebalances each back to the clients' target mix in an effort to optimize returns for their intended level of risk. After taking tax implications and trading costs into consideration, we rebalance when dividends from ETFs accrue, a deposit or withdrawal has been made or if movements in their relative allocations justify a change. Using cash inflows to buy underweighted asset classes is a smart rebalancing technique to minimize tax consequences and trading costs. We employ threshold based rebalancing, instead of time based rebalancing, to take advantage of market movements.

An analysis performed by David Swensen, Chief Investment Officer at Yale University, found rebalanced portfolios earned an average of 0.4% more per year, with less risk, over 10 years, than portfolios that were not rebalanced (Swensen, 2005). An analysis by Burt Malkiel and Charley Ellis found similar results over a different 10 year period (Malkiel & Ellis, 2013). More generally, rebalancing will usually reduce risk over time, but not necessarily increase returns. Rebalancing tends to increase returns in more volatile markets.

It is important to note that a client's asset allocation will typically need to be adjusted over time as her investment goals and risk tolerance may change. Wealthfront recommends our clients review their investment plans in detail every three to five years to determine whether their risk tolerance and target allocation should be updated. We also remind our clients on a quarterly basis to keep us informed of any such changes.

Conclusion

Wealthfront combines the judgment of its world-class investment team with state of the art optimization tools to identify efficient portfolios. We strive to deliver the maximum net-of-fee, after-tax, real investment return for each client's particular tolerance for risk. This means we will continue to look for meaningful ways to improve our investment methodology in the future while continuously

monitoring and periodically rebalancing our clients' portfolios to maximize returns while maintaining their calculated risk tolerance. We believe following this process will lead to outstanding long-term financial outcomes for our clients.

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Wealthfront presents the information starting in 1997, which is the earliest date that necessary data is available for all eleven of the asset classes being used. We have not made any additional calculations to account for the periodic rebalancing, which we use as part of the allocation plan, nor have we deducted other expenses. For our calculations, we use the following: U.S. stock (Russell 3000 Total Return Index), Foreign stock (MSCI EAFE Total Return Index), Emerging market stock (MSCI Emerging Markets Total Return Index), Dividend growth stock (Dow Jones Select Dividend Total Return Index), Real Estate (NAREIT North America Index), Natural Resources (XLE-S&P Energy Select Sector Index), TIPS (Barclays Capital U.S. TIPS Index), U.S. government bond (Barclays Capital U.S. Aggregate Bond Index), Corporate bond (Vanguard U.S. Intermediate-term Corporate Bond Mutual Fund Total Return), Municipal bond (Vanguard Intermediate-term Municipal Bond Mutual Fund Total Return), and Emerging market bond (GMO Emerging Market Bond Mutual Fund Total Return). Comparisons to indices are provided for illustrative purposes only.

Wealthfront's service was not available to investors during the time period shown. The choices made by Wealthfront to use certain indices may affect the performance calculations, and different choices would result in different performance estimates. The information is only an indication of the general performance of one type of allocation plan during the time period, and other allocation plans, based on different risk profile information, could have also been selected for comparison. No index is directly comparable to the performance of an asset class. Various strategies and assumptions may affect performance, such as ETF selection, ETF tracking error and expenses, and rebalancing of allocations.

The use of a different rebalancing plan could create different results. The deduction of expenses could create different results.

Past performance is no guarantee of future results, and any hypothetical returns, expected returns, or probability projections may not reflect actual future performance. Actual investors on Wealthfront may experience different results from the results shown. There is a potential for loss as well as gain that is not reflected in the hypothetical information portrayed. **The performance results shown do not represent the results of actual trading using client assets but were achieved by means of the retroactive application of a model designed with the benefit of hindsight.**

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